

1 Problem

- Top chess and shogi engines combined handcrafted evaluation, domain-specific search and decades of expert refinements.
- Can one learning algorithm master chess, shogi and Go from random play using only the rules?

2 Neural guide

ONE NETWORK, TWO OUTPUTS

- A residual network $f_{\theta}(s)$ outputs move priors p and a scalar value v predicting the eventual winner.
- The same architecture replaces handcrafted evaluation and rollout policies across all three games.

3 Search

MCTS IMPROVES THE POLICY

- Tree search selects moves using visit counts, priors and value estimates, then returns an improved policy π .
- Root noise and temperature encourage diverse early exploration; no random rollout to terminal states is needed.

4 Self-play learning

- $\text{loss} = (z-v)^2 - \pi^T \log(p) + c ||\theta||^2$
- Self-play stores states, MCTS policies and final outcomes; updated networks then generate stronger future games.

5 Evidence

- From tabula rasa, AlphaZero surpasses world-champion programs in chess, shogi and Go within 24 hours.
- In reported 100-game matches it scores 28-0-72 versus Stockfish, 90-8-2 versus Elmo, and 60-40 versus AlphaGo Zero.

6 Meaning & limits

- The work shows a common search-and-learning loop can replace large bodies of human domain knowledge in perfect-information games.
- It assumes exact rules and a cheap simulator, uses very large compute and does not directly solve stochastic real-world control.

ALPHAZERO

Mastering Chess and Shogi by Self-Play

Silver et al. | arXiv:1712.01815v1

Starting only from game rules, combine neural policy-value prediction with Monte Carlo tree search and repeatedly improve through self-play.

self-playMCTSpolicy-value network